

Kimchheang Chhea, Ph.D.

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🌐 Online Portfolio

🎓 Google Scholar

🔗 ORCID

Education

Ph.D. in AI/IoT/Big Data-based New Energy Industry

August 2019 - August 2025

Chung-Ang University, *Seoul, Korea*

Thesis: “Joint Time-average EE and PAoI Fairness Maximization for ISAC-UAV Using Attention-MAPPO”

Advisor: Prof. Jung-Ryun Lee

B. Eng. Electronics and Telecommunication

September 2014 - June 2019

Institute of Technology of Cambodia, *Phnom Penh, Cambodia*

B. Ed. Teaching English as a Foreign Language

Institute of Foreign Languages, *Phnom Penh, Cambodia*

September 2015 - June 2019

Experiences

Flynix Inc., Seoul, Korea

September 2025 - June 2026

Research Engineer

- Led a government research grant proposal for AI-based networks and smart factory enhancement.
- Established strategic partnerships for vertiport construction and Urban Air Mobility (UAM) integration.
- Prepared the final report for a government-funded project (USD 158,500) on the development of an interactive system for structural integrity and lifespan monitoring in aerospace applications.

Academic Service

Journal Reviewer

Elsevier Computer Networks

2025 – Present

IEEE Transactions on Cognitive Communications and Networking

2026 – Present

IEEE Transactions on Mobile Computing

2026 – Present

Scientific Reports (Nature Portfolio)

2026 – Present

Chung-Ang University, Seoul, Korea

August 2019 - August 2025

Research Assistant:

- Designed experimental frameworks contributing to 9 published and 4 in-progress journal papers.
- Supervised undergraduate and master’s students in research methodology and technical writing.
- Led seminar organization and developed structured study plans to enhance lab research output.
- Prepared project proposals including experimental designs, preliminary results, and research objectives.
- Managed laboratory administrative operations to ensure smooth day-to-day research workflows.

CamGSM Co. Ltd, Phnom Penh, Cambodia

March 2019 - July 2019

Intern: Planning fiber-optic transport infrastructure for IP networks

Awards

Graduate School Alumni Association President's Award

August 2025

CAU Young Scientist Scholarship (CAYSS)

August 2019 - June 2023

Technical Skills

1. OS: Windows, Linux
2. Programming languages: Python, MATLAB
3. Machine learning/Deep learning framework: Pytorch, TensorFlow, Numpy, Pandas with a focus on power/mode-selection/resource-allocation/secrecy optimization for energy efficiency in space-air-ground-integrated/device-to-device/cellular-underlay/wireless-body-area networks.
4. Languages: Khmer (Native), English (Professional), Korean (Intermediate)

Publications

1. **K. Chhea** and J.-R. Lee, "Enhancing Fairness in Energy Efficiency and Peak Age of Information in Low Earth Orbit-Aided Unmanned Aerial Vehicle-Integrated Sensing and Communication Networks: Attention-Aided Multi-Agent Proximal Policy Optimization Approach," *Engineering Applications of Artificial Intelligence*, vol. 181, Art. no. 115492, 2026, DOI: 10.1016/j.engappai.2026.115492.
2. **K. Chhea** and J. -R. Lee, "Joint Optimization of Secrecy Rate and Energy Consumption for RSMA in SAGIN via MADRL," *IEEE Transactions on Vehicular Technology*, vol. 75, no. 2, pp. 2983-2995, Feb. 2026, DOI: 10.1109/TVT.2025.3598635.
3. **K. Chhea** and J. -R. Lee, "Computing Efficiency and Latency Optimization in Intelligent Reflecting Surface-Enhanced Device-to-Device Mobile Edge Computing via Lagrange-Dual Deep Learning," *Journal of King Saud University Computer and Information Sciences*, vol. 38, no. 2, p. 59, Jan. 2026, DOI: 10.1007/s44443-025-00427-2.
4. S. Meng, **K. Chhea**, S. Muy and J. -R. Lee, "Deep Reinforcement Learning and Metaheuristic Approaches to Maximize Downlink Sum-Rate for IoT Systems in NOMA-based SAGIN," *Romanian Journal of Information Science and Technology*, vol. 28, no. 4, pp. 327-340, 2025, DOI: 10.59277/ROMJIST.2025.4.02.
5. **K. Chhea** and J. -R. Lee, "Enhancing Secure Energy Efficiency of SWIPT IoT Network Considering IRS and Artificial-Noise: A Deep Learning Approach," *IEEE Access*, vol. 13, pp. 63942-63951, 2025, DOI: 10.1109/ACCESS.2025.3559981.
6. **K. Chhea**, S. Muy, and J. -R. Lee, "Energy Efficiency Optimization in Intelligent Reflecting Surface-aided UAV Wireless Power Transfer Networks Using DRL," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 4, pp. 6599-6609, April 2025, DOI:10.1109/TVT.2024.3519591.
7. V. That, **K. Chhea**, and J. -R. Lee, "Optimizing Energy Consumption and Latency in IoT Through Edge Computing in Air-Ground Integrated Network With Deep Reinforcement Learning," *IEEE Open Journal of Vehicular Technology*, vol. 6, pp. 412-425, 2025, DOI:10.1109/OJVT.2024.3507288.

8. **K. Chhea**, D. Ron, and J. -R. Lee, “Weighted De-Synchronization Based Resource Allocation in Wireless Networks,” *Computer, Material & Continua*, vol. 75, no. 1, pp. 1815–1826, 2023. DOI:10.32604/cmc.2023.032376.
9. **K. Chhea** and J.-R. Lee, “Energy-efficient full-duplex D2D for SWIPT-empowered underlay cellular networks using a deep neural network,” *Computer Networks*, vol. 217, p. 109324, Nov. 2022. DOI:10.1016/j.comnet.2022.109324.
10. **K. Chhea**, D. Ron and J. -R. Lee, “Application of Kuramoto Model to Transmission Power Control in Wireless Body Area Networks,” *IEEE Access*, vol. 8, pp. 213531-213540, 2020, DOI:10.1109/ACCESS.2020.3040430.

Under Review:

11. **K. Chhea** and J. -R. Lee, “Joint Age of Information and Energy Consumption Optimization in SAGIN Using MADDPG.”
12. **K. Chhea** and J. -R. Lee, “Two-Tier Energy Efficiency Optimization and Trade-off in SAGIN.”
13. **K. Chhea** and J. -R. Lee, “Interference Mitigation in Space-Air-Ground Integrated Networks with an IRS-equipped HAP.”

International Conferences

1. **K. Chhea**, S. Meng, and J.-R. Lee, “Intelligent Surface-Assisted UAV Networks: A DRL Approach to Energy Efficiency,” *2025 International Conference on Information Networking (ICOIN)*, Jan. 2025, pp. 363-368, DOI: 10.1109/ICOIN63865.2025.10993013.
2. **K. Chhea**, S. Meng, and J.-R. Lee, “Enhancing Energy Efficiency in Underlay Cellular Networks: Leveraging Deep Learning for Full-Duplex SWIPT-based D2D Communications,” *2024 International Conference on Information Networking (ICOIN)*, Jan. 2024, pp. 138-146, DOI: 10.1109/ICOIN59985.2024.10572131.
3. **K. Chhea**, S. Muy, and J.-R. Lee, “Kuramoto-Inspired Wireless Resource Allocation for Weighted Networks,” *2023 International Conference on Information Networking (ICOIN)*, Jan. 2023, DOI: 10.1109/icoi56518.2023.10048922.

References

Jung-Ryun Lee, Ph.D., *Senior Member IEEE*

Professor

School of Electrical and Electronics Engineering

Director, Autonomous Intelligent Networks and Systems Laboratory

Head of Department, Intelligent Energy and Industry

Chung-Ang University

Email: jrlee@cau.ac.kr